EXPANDING MAPS: CLT

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1. Some questions

We have seen that, given a map $T \in \mathcal{C}^2(\mathbb{T}^1, \mathbb{T}^1)$, $|DT| \geq \nu > 1$, there exists a unique measure μ absolutely continuous with respect to Lebesgue (with density $h \in W^{1,2} \subset \mathcal{C}^0$) and the Dynamical System $(\mathbb{T}^1, T, \mu)$ is mixing. In particular this implies that, for each $f \in L^1$,

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} f\circ T^k = \mu(f) \quad \mu-\text{a.s.}$$

Yet, in any real experiment, n will be finite, maybe rather large, but always finite. It is then essential to have concrete estimates on how fast the limit is reached.

In particular a first question may be the following: given $f \in W^{1,2}$, $n \in \mathbb{N}$ and $a \in \mathbb{R}_+$ let

$$A_{a,n}(f) := \left\{ x \in \mathbb{T}^1 \;:\; \left| \frac{1}{n}\sum_{k=0}^{n-1} f\circ T^k(x) - \mu(f) \right| \geq a\mu(|f|) \right\}. \tag{1.1}$$

Question 1. *How large is $\mu(A_{a,n})$?*

Note that we can write $\frac{1}{n}\sum_{k=0}^{n-1} f\circ T^k(x) - \mu(f) = \frac{1}{n}\sum_{k=0}^{n-1} \hat f\circ T^k(x)$ where $\hat f := f - \mu(f)$. So we can reduce the question to the study of zero average function. A more refined question could be.

Question 2. *Does it exists a sequence $\{c_n\}$ such that*

$$\frac{1}{c_n}\sum_{k=0}^{n-1} \hat f\circ T^k(x)$$

converges in some sense to a non zero object?

1.1. Large deviations–well, half of it.

Note that it suffices to study the set

$$A^+_{a,n}(f) := \left\{ x \in \mathbb{T}^1 \;:\; \frac{1}{n}\sum_{k=0}^{n-1} f\circ T^k(x) - \mu(f + a|f|) \geq 0 \right\}.$$

since $A_{a,n}(f) = A^+_{a,n}(f) \cap A^+_{a,n}(-f)$. On the other hand, setting $f_a := f - \mu(f + a|f|)$ holds

$$m(A^+_{a,n}(f)) = \mu(\{x \;:\; e^{\lambda\sum_{k=0}^{n-1} f_a\circ T^k(x)} \geq 1\}) \leq \mu(e^{\lambda\sum_{k=0}^{n-1} f_a\circ T^k}) = m(he^{\lambda\sum_{k=0}^{n-1} f_a\circ T^k}).$$

Then

$$\mu(A^+_{a,n}(f)) \leq m(\mathcal{L}^n_\lambda h) \tag{1.2}$$

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where we have defined the operator $\mathcal{L}_\lambda g := \mathcal{L}(e^{\lambda f_a} g)$, $\mathcal{L}$ being the Transfer operator of the map T.

Since f_a is bounded it is easy to see that $\mathcal{L}_\lambda$ is a well defined operator on $W^{1,2}$.

The basic idea to accomplish the wanted estimate is to show that the spectral radius of $\mathcal{L}_\lambda$ is strictly smaller than one. Since we are interested in λ small we will try to apply perturbation theory viewing $\mathcal{L}_\lambda$ as a perturbation of $\mathcal{L}$. Since $\mathcal{L}_\lambda g = \sum_{n=0}^\infty \mathcal{L}(\frac{\lambda^n f_a^n g}{n!})$ it is clear that $\mathcal{L}_\lambda$ depends analytically from λ and we can thus apply the usual perturbation theory for operators (see [1]). Accordingly, for λ small enough, the maximal eigenvalue is close to one and the associated eigenspace is one dimensional. Hence, there exists $h_\lambda \in W^{1,2}$ and $\ell_\lambda \in W^{1,2\prime}$, both analytic in λ, such that the project on the maximal eigenvalue of $\mathcal{L}_\lambda$ reads $\Pi_\lambda(h) = h_\lambda \ell_\lambda(h)$. Obviously

$$\mathcal{L}_\lambda h_\lambda = \alpha_\lambda h_\lambda, \tag{1.3}$$

and $\alpha_0 = 1$, $h_0 = h$ and $\ell_0 = m$. Notice that h_λ and ℓ_λ are not uniquely defined: by $\Pi_\lambda^2 = \Pi_\lambda$ follows $\ell_\lambda(h_\lambda) = 1$ but one normalization can be chosen freely, let us choose $m(h_\lambda) = 1$. All the above discussion is summarize by the following Lemma.

Lemma 1.1. *There exists constants $C_1, C_2 > 0$ and $\rho > 0$ such that, for $\lambda \leq C_1 |f_a|_\infty^{-1}$, $\mathcal{L}_\lambda = \alpha_\lambda \Pi_\lambda + Q_\lambda$, $\Pi_\lambda Q_\lambda = Q_\lambda \Pi_\lambda = 0$, $\|Q_\lambda^n\|_{W^{1,2}} \leq C_2 \rho^n$. Moreover everything is analytic in λ.*

In view of the above fact we can differentiate (1.3) obtaining

$$\mathcal{L}_\lambda' h_\lambda + \mathcal{L}_\lambda h_\lambda' = \alpha_\lambda' h_\lambda + \alpha_\lambda h_\lambda' \; ; \quad m(h_\lambda') = 0. \tag{1.4}$$

Integrating with respect to m yields

$$\alpha_\lambda' = m(\mathcal{L}_\lambda(f_a h_\lambda)) + m(\mathcal{L}_\lambda h_\lambda').$$

Thus $\alpha_0' = \mu(f_a) = -a\mu(|f|) < 0$. This means that we can choose λ such that the norm of $\mathcal{L}_\lambda^n$ is strictly smaller than one, yet to know how small we can take it, it is necessary to investigate the second derivative of α_λ. Taking the derivative of (1.4), integrating with respect to m and setting $\lambda = 0$ yields

$$\alpha_0'' = m(h\, f_a^2) + 2m(f_a\, h_0').$$

On the other hand, setting $\Pi g := \Pi_0 g = h\, m(g)$, (1.4) implies

$$(\mathbf{Id} - \mathcal{L}) h_0' = \mathcal{L}(f_a h - \mu(f_a) h) = \mathcal{L}(\mathbf{Id} - \Pi)(f_a h).$$

On the other hand, setting $\hat{\mathcal{L}} := \mathcal{L}(\mathbf{Id} - \Pi)$, Lemma 1.1 implies that the spectral radius of $\hat{\mathcal{L}}$ is smaller than ρ, hence

$$h_0' = (\mathbf{Id} - \hat{\mathcal{L}})^{-1} \hat{\mathcal{L}}(f_a h), \tag{1.5}$$

and

$$\alpha_0'' = \mu(f_a^2) + 2m(f_a (\mathbf{Id} - \hat{\mathcal{L}})^{-1} \hat{\mathcal{L}}(f_a h)). \tag{1.6}$$

Since $\alpha_\lambda = 1 - \alpha_0' \lambda + \frac{1}{2} \alpha_0'' \lambda^2 + \mathcal{O}(\lambda^3)$, it follows that the situation is drastically different if α_0'' is positive or negative. Looking at (1.6) the sign is far from evident, yet a careful analysis shows that the sign is often positive.

Lemma 1.2. *Either $\alpha_0'' \geq C > 0$, with C independent on a, or, for each periodic orbit $\{x_i = T^i x_0\}_{i=0}^{n-1}$, it holds true $\sum_{i=0}^{n-1} \hat{f}(x_i) = 0$.*

Proof. First of all $(\mathbf{Id}-\hat{\mathcal{L}})^{-1}\hat{\mathcal{L}}g = (\mathbf{Id}-\hat{\mathcal{L}})^{-1}\hat{\mathcal{L}}\hat{f} \in \{g \in W^{1,2} \;:\; m(g)=0\}$, thus $\alpha_0'' = \mu(\hat{f}^2)+a^2\mu(|f|)^2+2m(\hat{f}(\mathbf{Id}-\hat{\mathcal{L}})^{-1}\hat{\mathcal{L}}(\hat{f}\,h)) \geq \mu(\hat{f}^2)+2m(\hat{f}(\mathbf{Id}-\hat{\mathcal{L}})^{-1}\hat{\mathcal{L}}(\hat{f}\,h))$. Next consider the following

$$
\begin{aligned}
0 \leq \mu\left(\left[\frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\hat{f}\circ T^k\right]^2\right) &= \frac{1}{n}\sum_{k,j=0}^{n-1}\mu(\hat{f}\circ T^k\hat{f}\circ T^j)\\
&= \mu(\hat{f}^2)+\frac{2}{n}\sum_{j=0}^{n-1}\sum_{k=j+1}^{n-1}\mu(\hat{f}\hat{f}\circ T^{k-j}) = \mu(\hat{f}^2)+\frac{2}{n}\sum_{k=1}^{n-1}\sum_{j=1}^{k}m(\hat{f}\hat{\mathcal{L}}^j(\hat{f}\,h))\\
&= \mu(\hat{f}^2)+2\sum_{j=1}^{n-1}\frac{n-j}{n}m(\hat{f}\hat{\mathcal{L}}^j(\hat{f}\,h)).
\end{aligned}
$$

Accordingly,

$$
\begin{aligned}
0 \leq \sigma^2 := \lim_{n\to\infty}\mu\left(\left[\frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\hat{f}\circ T^k\right]^2\right) &= \mu(\hat{f}^2)+2\sum_{j=1}^{\infty}m(\hat{f}\hat{\mathcal{L}}^j(\hat{f}\,h))\\
&= \mu(\hat{f}^2)+2m(\hat{f}(\mathbf{Id}-\hat{\mathcal{L}})^{-1}\hat{\mathcal{L}}(\hat{f}\,h)).
\end{aligned}
$$

Clearly, if $\sigma>0$ the lemma is proven, thus we need only to analyze the case $\sigma=0$.

If $\sigma=0$, we have

$$
\begin{aligned}
\mu\left(\left[\sum_{k=0}^{n-1}\hat{f}\circ T^k\right]^2\right) &= n\left[\mu(\hat{f}^2)+2\sum_{j=1}^{n-1}\frac{n-j}{n}m(\hat{f}\hat{\mathcal{L}}^j(\hat{f}\,h))\right]\\
&= -2n\sum_{j=n}^{\infty}m(\hat{f}\hat{\mathcal{L}}^j(\hat{f}\,h))-2\sum_{j=1}^{n-1}jm(\hat{f}\hat{\mathcal{L}}^j(\hat{f}\,h))\\
&\leq C_3\left[n\rho^n+\sum_{j=0}^{\infty}j\rho^j\right]|\hat{f}|_{L^2}\|\hat{f}\|_{W^{1,2}} \leq C_4|\hat{f}|_{L^2}\|\hat{f}\|_{W^{1,2}}
\end{aligned}
$$

Accordingly, the sequence $\sum_{k=0}^{n-1}\hat{f}\circ T^k$ is bounded in L^2 and hence weakly compact. Let $\sum_{k=0}^{n_j-1}\hat{f}\circ T^k$ a weakly convergent subsequence, that is there exists $\phi\in L^2$ such that for each $\varphi\in L^2$ holds

$$
\lim_{j\to\infty}\int\varphi\sum_{k=0}^{n_j-1}\hat{f}\circ T^k = \int\varphi\phi.
$$

It follows that, for each $\varphi\in W^{1,2}$,

$$
\begin{aligned}
\int\varphi[\hat{f}-\phi+\phi\circ T] &= m(\varphi\hat{f})+\lim_{j\to\infty}\sum_{k=0}^{n_j-1}\int\varphi\hat{f}\circ T^{k+1}-\int\varphi\hat{f}\circ T^k\\
&= \lim_{j\to\infty}\int\varphi\hat{f}\circ T^{n_j} = m(\varphi)\mu(\hat{f}) = 0
\end{aligned}
$$

And, since $W^{1,2}$ is dense in L^2, it follows

$$
\hat{f} = \phi-\phi\circ T. \tag{1.7}
$$

A function with the above property is called a *coboundary*, in this case an L^2 coboundary since we know only that $\phi \in L^2$. In fact, this it is not not enough to conclude the Lemma: we need to show, at least, that $\phi \in \mathcal{C}^0$.

First of all notice that, since for each $\beta \in \mathbb{R}$ we have $\hat{f} = \phi + \beta - (\phi + \beta) \circ T$, we can assume without loss of generality $\mu(\phi) = 0$. But them

$$\hat{\mathcal{L}} \hat{f}\, h = \mathcal{L} \hat{f}\, h = \mathcal{L}\phi\, h - \phi\, h = \hat{\mathcal{L}}\phi\, h - \phi\, h = -(\mathbf{Id} - \hat{\mathcal{L}})\phi\, h.$$

Hence[1]

$$\phi = h^{-1}(\mathbf{Id} - \hat{\mathcal{L}})^{-1}\hat{\mathcal{L}}(\hat{f}\, h) \in W^{1,2}$$

That is $\hat{f}$ is a continuous coboundary. From this the lemma follows immediately. $\square$

Accordingly, $\alpha_0'' = \sigma^2 + a^2\beta^2$, $\beta := \mu(|f|)$. The minimum (of the quadratic part) is thus achieved by choosing $\lambda_- = \frac{a\beta}{\sigma^2 + a^2\beta^2}$ which, remembering (1.2), allows to conclude

$$m(A^+_{a,n}(f)) \leq m(\mathcal{L}^n_{\lambda_-} h) \leq C e^{-\frac{a^2\beta^2}{2\sigma^2} n + \mathcal{O}(a^3 n \|f\|_{W^{1,2}})}.$$

Since similar arguments hold for the set $A^+_{a,n}(-f)$, it follows that we have an exponentially small probability to observe a deviation from the average. In particular we have the following weaker form of Birkhoff theorem.

Lemma 1.3. *For each $f \in W^{1,2}$, holds*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k(x) = m(f), \quad m - a.e.$$

Proof. Notice that, for each $f \in W^{1,2}$, holds $\sum_{n=0}^{\infty} m(A^+_{\delta,n}(f)) < \infty$ for each $\delta > 0$. The statement follows then by a simple application of Borel-Cantelli. $\square$

1.2. **The Central Limit Theorem.** We can now address the second question we have posed. From the above discussion is clear that we must chose $c_n = \sqrt{n}$.

Let $f \in W^{1,2}$ and set $\hat{f} := f - \mu(f)$, then

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \hat{f} \circ T^k(x) = 0 \quad m - \text{a.e.}$$

Let us set $\Psi_n := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \hat{f} \circ T^k$. We can consider Ψ_n a random variable with distribution $F_n(t) := mu(\{x \; : \; \Psi_n(x) \leq t\})$. It is well know that, for each

[1] Indeed, for each $\varphi \in L^2$,

$$\begin{aligned}\int \varphi(\mathbf{Id} - \hat{\mathcal{L}})^{-1}(\mathbf{Id} - \hat{\mathcal{L}})(h\phi) &= \int \varphi\phi h - \lim_{n\to\infty} \int \varphi \mathcal{L}^n h\phi \\ &= \int \varphi\phi h - \lim_{n\to\infty} \int \varphi \circ T^n h\phi = \int \varphi\phi\end{aligned}$$

where we have using the mixing of the system.

continuous function g holds[2]

$$\mu(g(\Psi_n)) = \int_{\mathbb{R}} g(t) dF_n(t)$$

where the integral is a Riemann-Stieltjes integral. It is thus clear that if we can control the distribution F_n, we have a very sharp understanding of the probability to have small deviations (of order $\sqrt{n}$) from the limit. From the work in the previous section it follows that there exists $\delta > 0$ such that, for each $|\lambda| \leq \delta\sqrt{n}$,

$$\begin{aligned}\varphi_n(\lambda) := \mu(e^{i\lambda\Psi_n}) = \mu(\mathcal{L}^n_{i\lambda/\sqrt{n}} h) &= (1 - \frac{\sigma^2\lambda^2}{2n} + \mathcal{O}(\lambda^3 n^{-\frac{3}{2}} + \rho^n)\|f\|_{W^{1,2}})^n \\ &= e^{-\frac{\sigma^2\lambda^2}{2}}(1 + \mathcal{O}(\lambda^3 n^{-\frac{1}{2}} + n\rho^n)\|f\|_{W^{1,2}}).\end{aligned} \tag{1.8}$$

The above quantity is called *characteristic function* of the random variable and determine the distribution via the formula

$$F_n(b) - F_n(a) = \lim_{\Lambda\to\infty} \frac{1}{2\pi}\int_{-\Lambda}^{\Lambda} \frac{e^{-ia\lambda} - e^{-ibt}}{i\lambda}\varphi_n(\lambda) d\lambda,$$

as can be seen in any basic book of probability theory.[3]

Formula (1.8) means in particular that

$$\lim_{n\to\infty} m(e^{\lambda\Psi_n}) = e^{-\frac{\sigma^2\lambda^2}{2}} =: \varphi(\lambda).$$

What can we infer out of the above facts? First of all a simple computation shows that

$$g(t) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-it\lambda}\varphi(\lambda) d\lambda = \frac{1}{\sqrt{\pi}\sigma} e^{-\frac{t^2}{2\sigma^2}}$$

a random variable with such a density is called a Gaussian random variable with zero average and variance σ. Accordingly, formula (1.8) can be interpreted by saying that there exists a Gaussian random variable G such that

$$\frac{1}{n}\sum_{k=0}^{n-1} \hat{f}\circ T^k \sim \frac{1}{\sqrt{n}} G(1 + \mathcal{O}(n^{-\frac{1}{2}}))$$

in distribution. But what does this means concretely. Actual estimates are made difficult by the fact that the distribution under study no not necessarily have a density, thus we are Fourier transforming function that behave quite badly at infinity. To overcome such a problem we can smoothen the quantities involved.

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[2] If $g \in \mathcal{C}^1_0$, then

$$\int_{\mathbb{R}} g dF_n = -\int_{\mathbb{R}} F_n(t) g'(t) dt = -\int_{\mathbb{R}} dt \int_{\mathbb{T}^1} dx \chi_{\{z\,:\,\Psi_n(z)\leq t\}}(x) g'(t).$$

Applying Fubini yields

$$\int_{\mathbb{R}} g dF_n = -\int_{\mathbb{T}^1} dx \int_{\mathbb{R}} dt \chi_{\{z\,:\,\Psi_n(z)\leq t\}}(x) g'(t) = -\int_{\mathbb{T}^1} dx \int_{\Psi_n(x)}^{\infty} g'(t) dt = \int_{\mathbb{T}^1} dx g(\Psi_n(x)).$$

[3] In the case when there exists a density, that is an L^1 function f_n such that $F_n(b) - F_n(a) = \int_a^b f_n(t) dt$, then the formula above becomes simply

$$f_n(t) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-it\lambda}\varphi_n(\lambda) d\lambda,$$

and follows trivially by the inversion of the Fourier transform.

Let $j \in \mathcal{C}^\infty(\mathbb{R}, \mathbb{R}_+)$ such that $\int_{\mathbb{R}} j(t)dt = 1$, $j(t) = j(-t)$, and $j(t) = 0$ for all $|t| > 1$, for each $\varepsilon > 0$ defined then $j_\varepsilon(t) := \varepsilon^{-1} j(\varepsilon^{-1} t)$ and

$$F_{n,\varepsilon}(t) := \int_{\mathbb{R}} j_\varepsilon(t-s) F_n(s) ds. \tag{1.9}$$

A simple computation shows that, for each $a, b \in \mathbb{R}$, holds

$$F_n(b+\varepsilon) - F_n(a-\varepsilon) \geq F_{n,\varepsilon}(b) - F_{n,\varepsilon}(a) \geq F_n(b-\varepsilon) - F_n(a+\varepsilon)$$

that is: if the measurements have a precision worst than 2ε, then $F_{n,\varepsilon}$ is as good as F_n to describe the resulting statistics. On the other hand calling $\varphi_{n,\varepsilon}$ the characteristic function associated to $F_{n,\varepsilon}$, holds $\varphi_{n,\varepsilon}(\lambda) = \varphi_n(\lambda)\hat{j}(\varepsilon\lambda)$, where $\hat{j}$ is the Fourier transform of j. Since now $F_{n,\varepsilon}$ is the law of a smooth random variable it has a density $f_{n,\varepsilon}$ and

$$f_{n,\varepsilon}(t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\lambda t} \varphi_n(\lambda) \hat{j}(\varepsilon\lambda) d\lambda$$

since j is smooth it follows that there exists $C > 0$ such that $|\hat{j}(\lambda)| \leq C(1+\lambda^2)^{-2}$. We can finally use formula (1.8) to obtain a quantitative estimate

$$\begin{aligned}
f_{n,\varepsilon}(t) &= \frac{1}{2\pi} \int_{-\varepsilon\sqrt{n}}^{\varepsilon\sqrt{n}} e^{-i\lambda t} \varphi_n(\lambda) \hat{j}(\varepsilon\lambda) d\lambda + \mathcal{O}(\varepsilon^{-5} n^{-\frac{3}{2}}) \\
&= \frac{1}{2\pi} \int_{-\varepsilon\sqrt{n}}^{\varepsilon\sqrt{n}} e^{-i\lambda t} \varphi(\lambda) \hat{j}(\varepsilon\lambda) d\lambda + \mathcal{O}(\varepsilon^{-5} n^{-\frac{3}{2}} + n^{-\frac{1}{2}}) \\
&= g(t) + \mathcal{O}(\varepsilon + \varepsilon^{-5} n^{-\frac{3}{2}} + n^{-\frac{1}{2}}) = g(t) + \mathcal{O}(n^{-\frac{1}{2}})
\end{aligned}$$

provided we choose $n^{-\frac{1}{2}} \geq \varepsilon \geq n^{-5}$. Which, as announced, means that, if the precision of the instrument is compatible with the statistics, the typical fluctuations in measurements are of order $\frac{1}{\sqrt{n}}$ and Gaussian. This is well known by sperimentalists who routinely assume that the result of a measurement is distributed according to a Gaussian.[4]

References

[1] T.Kato, *Perturbation theory fro linear operators*, Springer, New York (1966).

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[4]Note however that our proof holds in a very special case that has little to do with a real experimental setting. To prove the analogous statement in for a realistic experiment is a completely different ball game.